

Markscheme

Mathematics: analysis and approaches
Practice question bank

117. (a)

A code consists of 2 letters (chosen from the 26-letter alphabet) followed by 4 digits (chosen from 0-9), with no repetition allowed within the letters, and no repetition allowed within the digits.

Find the number of possible codes.

letters: 26×25 *(M1)*

digits: $10 \times 9 \times 8 \times 7$ *(M1)*

multiply: 650×5040 *(M1)*

$= 3\,276\,000$ **A1**

[4 marks]